

were more active or our solutions detected more malicious files simply because of greater activity. It could be a combination of both. Either way, we have registered a noticeable increase in the number of new malicious files this year, and this will most likely continue going into 2021 as employees continue to work from home and countries implement different restrictions. However, if users take basic security precautions, they can significantly lower their risk of encountering them.”

Sonit Jain, CEO of GajShield Info-tech explains the year that was and what to look forward to, “2020 forced us through a complete revamp of enterprises’ work infrastructure. From working at a minimum on-premise capacity to complete remote working, we saw various challenges with these working conditions. Enterprises rushed the work from home setup and the adoption of cloud infrastructure that increased dependence on SaaS/cloud application, opening a larger challenge in ensuring data security. Negating these challenges can be difficult by holding on to the older security approach, the impact of which we saw in the recent data breaches faced by Spotify, Big Basket etc. This new work infrastructure demands for a more robust and purpose-built security solution with a newer approach to security, such as the Data Security Approach. It keeps data at the centre of all its initiatives, covers enterprise’s SaaS/Cloud application platforms and brings the remote/WFH users under the security infrastructure to prevent intentional/unintentional, internal and external data exploitation while protecting from network threats, keeping us ready for the future.”

Netwrix meanwhile has identified misconfiguration in cloud technologies as a potential source of future data breaches. According to security experts, one of the problems with cloud security is that the companies that buy the services are not entirely sure about who is responsible for the data and files in the service. Meanwhile, the vendors of the service put the onus on their customers to secure their data. To specifically address

this lack of clarity, Trend Micro has announced a cloud native security solution in December 2020, the first of its kind in the world.

Mark Nunnikhoven, Vice-President of Cloud Research for Trend Micro says, “Global organizations are increasingly looking to public cloud providers to drive IT agility, cost savings and business growth. But while the provider deals with security of the cloud, the customer is responsible for everything inside their cloud environment. This is a highly scalable, automated scanning tool that’s fast to deploy with no added infrastructure, allowing organizations to confidently store cloud files and data associated with their cloud applications.”

as its alleged spread via 5G, could be reemphasised via deepfake videos, for example wrongly showing politicians as conspirators. The pandemic, the resulting increase in people working from home, and higher reliance on online connectivity as well as the growing economic pressure, combined with uncertainty among people, are likely to feed into the effectiveness of the use of deepfakes to spread disinformation.”

BeyondTrust and Kaspersky have also warned of deepfake campaigns in their security forecasts. The technology is sophisticated enough that now users have to be wary that they are interacting with the real person, even over a live video stream. At the



Credit: RepresentUS

A still from the “Dictators” DeepFake campaign by RepresentUS, a non partisan organisation advocating electoral reforms in the US

## 13 DEEPFAKES GET PERSONAL

The use of deepfakes for spreading disinformation is going to become increasingly effective and improve in quality. While even the most basic tricks such as editing a video or audio file can work wonders while racing around the shady WhatsApp circuits, we will be consuming media without any realisation that they may be faked, and this is going to help the actors behind the disinformation campaigns.

Petr Somol, AI Research Director at Avast explains, “deepfakes will likely reach a quality next year where they can be actively used in disinformation campaigns. Conspiracy theories about the coronavirus, such

same time, deepfakes can be expected to go beyond video, and move into stuff such as voice messages on instant messaging services, or interactions on platforms such as Snapchat, Discord and Skype. Securelist by Kaspersky notes that it is possible for threat actors to use live and targeted communication for spear phishing attacks. What this potentially means is that you could get a video call from your harried boss from an unknown number asking for credentials to company servers, and you could end up handing them over based on a deepfaked call. In fact, exactly such an attack has taken place already where cybercriminals used AI to fake the voice of the CEO of a German firm and got the CEO of the UK wing of